# Cohen Bryant

Madison, WI | cohenbryant22@gmail.com | 828-750-2448 | LinkedIn | Professional Portfolio Link

## Professional Summary

Mechanical Engineering student at UW-Madison with hands-on experience in computational physics modeling, CAD/FEA analysis, and embedded systems development. Proven ability to translate engineering theory into working technical systems, from nuclear reactor transient simulations to autonomous hardware platforms. Seeking roles in aerospace, defense, or nuclear energy where physics-driven analysis and multidisciplinary technical skills drive impact.

## Education

**University of Wisconsin–Madison** Madison, WI
*Bachelor of Science in Mechanical Engineering.* *Expected May 2030*
Relevant Coursework: Calculus II/III, Statics, Introduction to Engineering Design, Computer-Aided Engineering (SolidWorks), Python/MATLAB Programming

## Skills

**CAD & Simulation:** SolidWorks (3D Assemblies, GD&T, Stress, Vibration & Aerodynamic FEA), FEA/CFD Principles
**Hardware & Electronics:** Arduino Nano ESP32, Motor Controllers, Sensors (Thermocouples, Load Cells), Microcontrollers, PCB Design, Electronic Schematics
**Fabrication & Machining:** 3D Printing (FDM/SLA), Lathe, Mill, Soldering, Hand Tools, Prototype Assembly
**Programming & Tools:** C++, Python, MATLAB, JavaScript, HTML/CSS, Git, Vercel, Arduino IDE

## Projects

**Transient Nuclear Reactor Xenon-135 Simulator https://xenon-project-six.vercel.app/**
*Creator & Developer*
- Self-initiated computational simulator built out of personal interest in nuclear energy, modeling transient Xenon-135 reactivity poisoning and reactor dead-out periods following thermal neutron flux step changes.
- Implemented differential decay kinetics for Iodine-135 and Xenon-135 in JavaScript to plot dynamic power shifts, peak poison concentrations, and override windows.
- Designed an interactive UI to evaluate reactor control constraints and steady-state equilibrium power transitions.

**Nuclear Systems Analytics Dashboard (NucInfo) https://nucinfo.vercel.app/**
*Developer & Researcher*
- Developed an interactive web analytics platform visualizing nuclear fission/fusion thermodynamics, 93% capacity factor metrics, and lifecycle carbon intensity (12 g CO2/kWh).

**Arduino ESP32 Hardware Controls (More In-depth Details on my Website)**
*Hardware & Control Lead*
- Programmed an Arduino Nano ESP32 in C++ to manage motor drive logic, speed control, and hardware signal processing for an autonomous vehicle platform.
- Modeled custom chassis and drivetrain components in SolidWorks, applying tight tolerances for 3D-printed drive sprockets and motor brackets.
- Debugged runtime C++ compilation errors, structured code directories, and documented wiring schematics to ensure hardware reliability.

**Moving Target Launcher & Trajectory Calculations**
*Mechanical Design & Analysis*
- Applied calculus and physics equations to model energy transfer, apex height, and trajectory distance for a ping pong ball launcher with an angle-adjustable base designed to engage a moving target.
- Machined structural support components and adjusted spring tension parameters to align physical test launch data within 5% of theoretical predictions.

**ME Connect Student Engineering Platform https://me-connect.vercel.app/**
*Creator & Web Architect*
- Architected and deployed a centralized networking platform to connect UW-Madison engineering undergraduates with faculty, mentorship portals, and course resources.

## Activities & Leadership

**AIAA / Liquid Rocketry / ANS (American Nuclear Society)** Aug 2026 – Present

*Active Member*

- Contribute to CAD modeling, subsystem fabrication, and test rig setup alongside upperclassmen engineers for upcoming competition builds.
- Contributions to a more sustainable source of energy and towards enacting a more powerful source of energy for the future, including fusion and fission technology, to find a viable end to market volatility and a greener future.

## Experience

**Certified Nursing Assistant** Aug 2024 – Jul 2026

*North Central Health Care*

- Coordinated patient care tasks with a multidisciplinary team in a high-acuity clinical environment, maintaining clear communication under time pressure.
- Adapted quickly to shifting patient needs and staff priorities, demonstrating reliability and composure in demanding workplace conditions.

## Certifications

- Microsoft Office Specialist Certificate, 2023
- MATHLAB Onramp (MathWorks) Certificate, 2026

Portfolio Website



LinkedIn Profile



Handshake Profile



Resume Digital

